BCA309T: Cryptography & Network Security

ACADEMIC SOLUTIONS MANUAL

Assignment 1: Foundations of Cryptography | Comprehensive Reference

Q1 (Short) Definition and Three Core Goals of Cryptography

Cryptography is the science and mathematical art of securing information and communications by transforming intelligible data into unintelligible formats, ensuring protection against unauthorized interception or tampering. Its three primary goals are: **(1) Confidentiality:** ensuring only authorized parties can access the message; **(2) Integrity:** guaranteeing data has not been altered, inserted, or deleted in transit; and **(3) Availability:** ensuring data and authentication systems remain accessible to authorized entities.

Q2 (Short) Plaintext vs. Ciphertext

Plaintext is the original, human-readable data before cryptographic transformation (e.g., the string `"TRANSFER $500"`). **Ciphertext** is the unreadable, scrambled output produced by applying an encryption algorithm under a specific key (e.g., `"9f8a3c8e71b2"`). Plaintext reveals semantic meaning immediately, whereas ciphertext is statistically random and incomprehensible without the corresponding decryption key.

Q3 (Long) Concept of Cryptography and Modern Real-Life Significance

Cryptography underpins modern cyberspace, enabling digital trust across open, untrusted networks like the Internet. Without cryptographic guarantees, digital systems collapse under identity theft, man-in-the-middle attacks, and eavesdropping. Key applications include:

- **Digital Banking & ATM Networks:** Protocols like EMV and TLS secure account credentials and transaction integrity, preventing fraudulent credit allocations and account hijackings.
- **End-to-End Encrypted Messaging:** Platforms like Signal and WhatsApp employ the Double Ratchet Algorithm to ensure no intermediate ISP, government, or server operator can inspect chat payloads.
- **E-Commerce (HTTPS/TLS):** Protects credit card numbers, authentication cookies, and order specifics during checkout against wiretapping and DNS spoofing.

Q4 (Short) Difference Between Encryption and Decryption

Encryption is the forward mathematical process $C = E_K(P)$ that transforms plaintext (P) into ciphertext (C) using an encryption key. **Decryption** is the inverse operation $P = D_K(C)$ that restores ciphertext back to original plaintext using the authorized decryption key. Encryption guarantees protection; decryption recovers accessibility.

Q5 (Long) Symmetric vs. Asymmetric Cryptography: Comparative Architecture

Metric	Symmetric Key Cryptography	Asymmetric (Public Key) Cryptography
Key Structure	Single shared secret key for encryption & decryption.	Key pair: Public key (encrypt/verify), Private key (decrypt/sign).
Computational Speed	Extremely fast; hardware-accelerated (e.g., AES-NI).	Computationally slow (~1000× slower due to modular arithmetic).
Key Exchange	Vulnerable; requires secure out-of-band channel.	Public keys can be freely published; solves key distribution.
Primary Use Cases	Bulk payload encryption, disk encryption, VPN tunnels.	Digital signatures, PKI certificates, key negotiation (TLS handshake).
Representative Algorithms	AES (128/256), DES, 3DES, ChaCha20.	RSA (2048/4096), ECC (ECDSA, Ed25519), Diffie-Hellman.

Q6 (Short) Hashing vs. Encryption

Hashing is a one-way mathematical function that maps arbitrary-length input to a fixed-size digest (e.g., SHA-256); it is irreversible by design and does not use a key. **Encryption** is a two-way transformation designed to be inverted using an authorized decryption key. Hashing verifies data integrity; encryption protects data confidentiality.

Q7 (Short) Challenge of Achieving Confidentiality, Integrity, and Non-Repudiation Jointly

Simultaneously achieving all three is challenging because each targets an orthogonal vulnerability: encryption provides confidentiality but does not inherently guarantee integrity or origin authenticity (an attacker can alter ciphertext blocks undetected without authenticated encryption like AES-GCM). Furthermore, symmetric methods cannot achieve non-repudiation because both parties possess the identical key. Coordinating digital signatures, PKI certificate chains, and key-management lifecycle introduces significant cryptographic overhead and protocol complexity.

Q8 (Long) Role of Cryptography in Network Security

Cryptography provides the foundational primitives required to secure distributed networked systems across four essential axes:

- **Confidentiality:** Prevents unauthorized packet eavesdropping on public transmission media. *Example:* IPsec ESP tunnels encrypting entire IP packets traversing branch offices.
- **Integrity:** Detects in-transit alteration, truncation, or bit flipping. *Example:* HMAC-SHA256 appended to data packets to guarantee packet content remains untouched.
- **Authentication:** Confirms the true identity of communicating endpoints. *Example:* X.509 digital certificates verifying the identity of an online banking server during TLS handshake.
- **Non-Repudiation:** Prevents an entity from denying the origin or submission of an action. *Example:* An RSA digital signature on a digitally signed financial transaction contract.

Q9 (Short) Network Security Protocols

1. SSL/TLS (Transport Layer Security): Secures application-layer protocols (HTTP, SMTP) by establishing an encrypted, authenticated session over TCP. **2. IPsec (Internet Protocol Security):** Operates at the Network Layer (OSI Layer 3) to authenticate and encrypt IP packet flows between gateways or hosts via AH and ESP.

Q10 & Q11 (Long) Evolution of Cryptography: Classical Era to Modern Computing

Cryptography evolved across three paradigm shifts driven by computational advances:

- **Classical Era (Manual & Linguistic):** Relied on substitution and transposition (Caesar, Vigenère). Security relied on algorithm secrecy (obscurity). These fell completely to statistical *frequency analysis* developed by Al-Kindi and Babbage.
- **Mechanical & WWII Era (Rotor Machines):** The German Enigma introduced polyalphabetic substitution via motorized mechanical rotors. Breaking Enigma at Bletchley Park via Turing's electromechanical "Bombe" marked the birth of automated cryptanalysis.
- **Modern Computing Standards:** Claude Shannon's 1949 treatise laid mathematical foundations (Confusion and Diffusion). IBM and NIST formalized the *Data Encryption Standard (DES, 1977)* using Feistel networks. As Moore's Law enabled 56-bit brute-force exhaustion (DES cracker EFF 1998), NIST standardized the *Advanced Encryption Standard (AES, 2001)* with 128/256-bit keys.
- **The Public-Key Revolution:** Solved the longstanding symmetric $O(n^2)$ key-distribution bottleneck through one-way trapdoor mathematical functions (RSA, ECC).

Q12 (Short) Introduction of Public-Key Cryptography

The concept of public-key cryptography was publicly introduced by **Whitfield Diffie and Martin Hellman** in their seminal 1976 paper, *"New Directions in Cryptography."* (Classified work was independently conducted earlier in 1969–1973 by James Ellis, Clifford Cocks, and Malcolm Williamson at Britain's GCHQ).